

Stratified drug-response prediction metrics (CPA family)

Metrics shown: Pearson r , Spearman ρ , and RMSE (mean $\pm$ s.d. across folds). Measured-profile rows use T1 continuous ElasticNet outcomes from measured-task CV. Best value per dataset is bold (higher Pearson/Spearman, lower RMSE). Comparison tables report paired fold differences versus Average effect and No effect baselines.

Table 1: Pooled fold-level metrics (CPA family; mean $\pm$ s.d. across folds; best per dataset in bold).

Dataset	Source	Model	Pearson r	Spearman ρ	RMSE
McFarland	Average effect	LFC	-0.002 $\pm$ 0.083	0.007 $\pm$ 0.057	0.178 $\pm$ 0.005
McFarland	Average effect	Post	0.011 $\pm$ 0.039	0.020 $\pm$ 0.036	0.176 $\pm$ 0.006
McFarland	CPA	LFC	0.521 $\pm$ 0.072	0.431 $\pm$ 0.035	0.150 $\pm$ 0.006
McFarland	CPA	Post	0.480 $\pm$ 0.064	0.399 $\pm$ 0.059	0.155 $\pm$ 0.006
McFarland	CPA End-to-End	Embedding	0.567 $\pm$ 0.074	0.517 $\pm$ 0.054	0.176 $\pm$ 0.013
McFarland	Measured	LFC	0.649 $\pm$ 0.031	0.552 $\pm$ 0.022	0.135 $\pm$ 0.007
McFarland	Measured	Post	0.573 $\pm$ 0.053	0.488 $\pm$ 0.031	0.147 $\pm$ 0.006
McFarland	Measured	Pre	0.092 $\pm$ 0.119	0.074 $\pm$ 0.088	0.175 $\pm$ 0.005
McFarland	Measured	Pre+SMILES	0.491 $\pm$ 0.059	0.442 $\pm$ 0.041	0.153 $\pm$ 0.008
McFarland	No effect	LFC	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.175 $\pm$ 0.007
McFarland	No effect	Post	0.009 $\pm$ 0.046	0.019 $\pm$ 0.052	0.176 $\pm$ 0.006
McFarland	PRnet	LFC	0.371 $\pm$ 0.034	0.334 $\pm$ 0.020	0.163 $\pm$ 0.007
McFarland	PRnet	Post	0.378 $\pm$ 0.138	0.375 $\pm$ 0.057	0.253 $\pm$ 0.218
McFarland	chemCPA	LFC	0.413 $\pm$ 0.074	0.380 $\pm$ 0.070	0.165 $\pm$ 0.002
McFarland	chemCPA	Post	0.200 $\pm$ 0.066	0.152 $\pm$ 0.042	0.172 $\pm$ 0.003
SciPlex3	Average effect	LFC	0.036 $\pm$ 0.055	0.034 $\pm$ 0.049	0.150 $\pm$ 0.029
SciPlex3	Average effect	Post	0.037 $\pm$ 0.056	0.034 $\pm$ 0.049	0.150 $\pm$ 0.029
SciPlex3	CPA	LFC	0.561 $\pm$ 0.108	0.409 $\pm$ 0.117	0.164 $\pm$ 0.015
SciPlex3	CPA	Post	0.587 $\pm$ 0.114	0.455 $\pm$ 0.059	0.162 $\pm$ 0.014
SciPlex3	CPA End-to-End	Embedding	0.430 $\pm$ 0.059	0.277 $\pm$ 0.183	0.231 $\pm$ 0.021
SciPlex3	Measured	LFC	0.705 $\pm$ 0.069	0.381 $\pm$ 0.088	0.139 $\pm$ 0.030
SciPlex3	Measured	Post	0.621 $\pm$ 0.114	0.390 $\pm$ 0.167	0.153 $\pm$ 0.038
SciPlex3	Measured	Pre	0.218 $\pm$ 0.071	0.076 $\pm$ 0.181	0.184 $\pm$ 0.023
SciPlex3	Measured	Pre+SMILES	0.245 $\pm$ 0.227	0.139 $\pm$ 0.147	0.181 $\pm$ 0.027
SciPlex3	No effect	LFC	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.150 $\pm$ 0.029
SciPlex3	No effect	Post	0.036 $\pm$ 0.055	0.034 $\pm$ 0.049	0.150 $\pm$ 0.029
SciPlex3	PRnet	LFC	0.474 $\pm$ 0.110	0.251 $\pm$ 0.068	0.170 $\pm$ 0.020
SciPlex3	PRnet	Post	0.496 $\pm$ 0.099	0.300 $\pm$ 0.091	0.168 $\pm$ 0.019
SciPlex3	chemCPA	LFC	0.406 $\pm$ 0.133	0.142 $\pm$ 0.040	0.175 $\pm$ 0.008
SciPlex3	chemCPA	Post	0.294 $\pm$ 0.188	0.137 $\pm$ 0.080	0.183 $\pm$ 0.014

Table 2: Paired fold comparisons vs baselines (pooled fold-level metrics; CPA family; best mean diff / lowest FDR p per dataset $\times$ metric in bold).

Dataset	Metric	Source	Model	Baseline	Mean diff [95% CI]	p (FDR)	n
McFarland	pearson	CPA	LFC	Average effect	0.522 [0.486, 0.558]	2.75e-05	5
McFarland	pearson	CPA	LFC	No effect	0.521 [0.431, 0.610]	0.000209	5
McFarland	pearson	CPA	Post	Average effect	0.469 [0.391, 0.547]	0.000209	5
McFarland	pearson	CPA	Post	No effect	0.471 [0.397, 0.545]	0.000209	5
McFarland	pearson	PRnet	LFC	Average effect	0.372 [0.262, 0.482]	0.00123	5
McFarland	pearson	PRnet	LFC	No effect	0.371 [0.328, 0.413]	0.000101	5
McFarland	pearson	PRnet	Post	Average effect	0.367 [0.198, 0.536]	0.00496	5
McFarland	pearson	PRnet	Post	No effect	0.369 [0.195, 0.543]	0.00496	5
McFarland	pearson	chemCPA	LFC	Average effect	0.414 [0.222, 0.606]	0.00496	5
McFarland	pearson	chemCPA	LFC	No effect	0.413 [0.320, 0.505]	0.000486	5
McFarland	pearson	chemCPA	Post	Average effect	0.189 [0.062, 0.316]	0.0159	5
McFarland	pearson	chemCPA	Post	No effect	0.192 [0.055, 0.328]	0.0176	5
McFarland	rmse	CPA	LFC	Average effect	-0.027 [-0.032, -0.022]	0.00129	5
McFarland	rmse	CPA	LFC	No effect	-0.025 [-0.034, -0.016]	0.00335	5
McFarland	rmse	CPA	Post	Average effect	-0.021 [-0.027, -0.016]	0.0013	5
McFarland	rmse	CPA	Post	No effect	-0.021 [-0.027, -0.016]	0.0013	5
McFarland	rmse	PRnet	LFC	Average effect	-0.014 [-0.019, -0.009]	0.00345	5
McFarland	rmse	PRnet	LFC	No effect	-0.012 [-0.015, -0.009]	0.0013	5
McFarland	rmse	PRnet	Post	Average effect	0.077 [-0.187, 0.341]	0.463	5
McFarland	rmse	PRnet	Post	No effect	0.077 [-0.187, 0.341]	0.463	5
McFarland	rmse	chemCPA	LFC	Average effect	-0.012 [-0.020, -0.005]	0.0165	5
McFarland	rmse	chemCPA	LFC	No effect	-0.010 [-0.019, -0.001]	0.0502	5
McFarland	rmse	chemCPA	Post	Average effect	-0.004 [-0.011, 0.003]	0.223	5
McFarland	rmse	chemCPA	Post	No effect	-0.004 [-0.011, 0.003]	0.223	5
McFarland	spearman	CPA	LFC	Average effect	0.425 [0.358, 0.492]	0.000243	5
McFarland	spearman	CPA	LFC	No effect	0.431 [0.388, 0.475]	6.07e-05	5
McFarland	spearman	CPA	Post	Average effect	0.378 [0.292, 0.465]	0.000454	5
McFarland	spearman	CPA	Post	No effect	0.380 [0.297, 0.462]	0.000454	5
McFarland	spearman	PRnet	LFC	Average effect	0.327 [0.249, 0.405]	0.000458	5
McFarland	spearman	PRnet	LFC	No effect	0.334 [0.309, 0.359]	3.76e-05	5
McFarland	spearman	PRnet	Post	Average effect	0.355 [0.281, 0.429]	0.000454	5
McFarland	spearman	PRnet	Post	No effect	0.356 [0.264, 0.448]	0.000567	5
McFarland	spearman	chemCPA	LFC	Average effect	0.373 [0.217, 0.529]	0.00323	5
McFarland	spearman	chemCPA	LFC	No effect	0.380 [0.293, 0.466]	0.000454	5
McFarland	spearman	chemCPA	Post	Average effect	0.131 [0.041, 0.222]	0.0173	5
McFarland	spearman	chemCPA	Post	No effect	0.133 [0.021, 0.244]	0.0297	5
SciPlex3	pearson	CPA	LFC	Average effect	0.525 [0.445, 0.605]	0.000256	5
SciPlex3	pearson	CPA	LFC	No effect	0.561 [0.427, 0.695]	0.00094	5
SciPlex3	pearson	CPA	Post	Average effect	0.550 [0.463, 0.637]	0.000256	5
SciPlex3	pearson	CPA	Post	No effect	0.551 [0.463, 0.638]	0.000256	5
SciPlex3	pearson	PRnet	LFC	Average effect	0.439 [0.291, 0.586]	0.00177	5
SciPlex3	pearson	PRnet	LFC	No effect	0.474 [0.338, 0.611]	0.00121	5
SciPlex3	pearson	PRnet	Post	Average effect	0.459 [0.324, 0.594]	0.00121	5
SciPlex3	pearson	PRnet	Post	No effect	0.459 [0.324, 0.595]	0.00121	5
SciPlex3	pearson	chemCPA	LFC	Average effect	0.370 [0.141, 0.598]	0.013	5
SciPlex3	pearson	chemCPA	LFC	No effect	0.406 [0.241, 0.571]	0.00322	5
SciPlex3	pearson	chemCPA	Post	Average effect	0.257 [-0.038, 0.552]	0.0729	5
SciPlex3	pearson	chemCPA	Post	No effect	0.257 [-0.038, 0.552]	0.0729	5
SciPlex3	rmse	CPA	LFC	Average effect	0.014 [-0.009, 0.038]	0.208	5
SciPlex3	rmse	CPA	LFC	No effect	0.014 [-0.010, 0.038]	0.208	5
SciPlex3	rmse	CPA	Post	Average effect	0.012 [-0.013, 0.037]	0.244	5
SciPlex3	rmse	CPA	Post	No effect	0.012 [-0.013, 0.037]	0.244	5
SciPlex3	rmse	PRnet	LFC	Average effect	0.020 [-0.009, 0.050]	0.196	5
SciPlex3	rmse	PRnet	LFC	No effect	0.020 [-0.009, 0.050]	0.196	5
SciPlex3	rmse	PRnet	Post	Average effect	0.018 [-0.008, 0.045]	0.196	5

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Table 2: Paired fold comparisons vs baselines (pooled fold-level metrics; CPA family; best mean diff / lowest FDR p per dataset $\times$ metric in bold). (continued)

Dataset	Metric	Source	Model	Baseline	Mean diff [95% CI]	p (FDR)	n
SciPlex3	rmse	PRnet	Post	No effect	0.018 [-0.008, 0.045]	0.196	5
SciPlex3	rmse	chemCPA	LFC	Average effect	0.025 [-0.006, 0.055]	0.196	5
SciPlex3	rmse	chemCPA	LFC	No effect	0.025 [-0.006, 0.056]	0.196	5
SciPlex3	rmse	chemCPA	Post	Average effect	0.033 [0.006, 0.059]	0.159	5
SciPlex3	rmse	chemCPA	Post	No effect	0.033 [0.006, 0.059]	0.159	5
SciPlex3	spearman	CPA	LFC	Average effect	0.375 [0.264, 0.485]	0.00284	5
SciPlex3	spearman	CPA	LFC	No effect	0.409 [0.264, 0.554]	0.00288	5
SciPlex3	spearman	CPA	Post	Average effect	0.421 [0.370, 0.473]	0.000136	5
SciPlex3	spearman	CPA	Post	No effect	0.421 [0.370, 0.473]	0.000136	5
SciPlex3	spearman	PRnet	LFC	Average effect	0.217 [0.093, 0.342]	0.0111	5
SciPlex3	spearman	PRnet	LFC	No effect	0.251 [0.167, 0.336]	0.00288	5
SciPlex3	spearman	PRnet	Post	Average effect	0.266 [0.142, 0.390]	0.00599	5
SciPlex3	spearman	PRnet	Post	No effect	0.266 [0.142, 0.390]	0.00599	5
SciPlex3	spearman	chemCPA	LFC	Average effect	0.107 [0.008, 0.207]	0.0482	5
SciPlex3	spearman	chemCPA	LFC	No effect	0.142 [0.092, 0.191]	0.00288	5
SciPlex3	spearman	chemCPA	Post	Average effect	0.103 [-0.051, 0.258]	0.138	5
SciPlex3	spearman	chemCPA	Post	No effect	0.103 [-0.051, 0.258]	0.138	5

Table 3: Within-drug metrics (averaged to fold level) (CPA family; mean $\pm$ s.d. across folds; best per dataset in bold).

Dataset	Source	Model	Pearson r	Spearman ρ	RMSE
McFarland	Average effect	LFC	0.052 $\pm$ 0.085	0.058 $\pm$ 0.081	0.160 $\pm$ 0.006
McFarland	Average effect	Post	0.047 $\pm$ 0.074	0.050 $\pm$ 0.070	0.159 $\pm$ 0.006
McFarland	CPA	LFC	0.282 $\pm$ 0.107	0.221 $\pm$ 0.080	0.140 $\pm$ 0.007
McFarland	CPA	Post	0.284 $\pm$ 0.120	0.223 $\pm$ 0.071	0.145 $\pm$ 0.005
McFarland	CPA End-to-End	Embedding	0.050 $\pm$ 0.066	0.059 $\pm$ 0.075	0.166 $\pm$ 0.012
McFarland	Measured	LFC	0.377 $\pm$ 0.054	0.290 $\pm$ 0.031	0.124 $\pm$ 0.009
McFarland	Measured	Post	0.329 $\pm$ 0.085	0.241 $\pm$ 0.039	0.136 $\pm$ 0.007
McFarland	Measured	Pre	0.114 $\pm$ 0.084	0.101 $\pm$ 0.066	0.159 $\pm$ 0.007
McFarland	Measured	Pre+SMILES	0.145 $\pm$ 0.067	0.142 $\pm$ 0.028	0.143 $\pm$ 0.012
McFarland	No effect	LFC	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.157 $\pm$ 0.008
McFarland	No effect	Post	0.045 $\pm$ 0.078	0.049 $\pm$ 0.069	0.159 $\pm$ 0.006
McFarland	PRnet	LFC	0.085 $\pm$ 0.053	0.072 $\pm$ 0.019	0.152 $\pm$ 0.009
McFarland	PRnet	Post	0.067 $\pm$ 0.071	0.057 $\pm$ 0.072	0.213 $\pm$ 0.157
McFarland	chemCPA	LFC	0.297 $\pm$ 0.060	0.281 $\pm$ 0.066	0.150 $\pm$ 0.003
McFarland	chemCPA	Post	0.032 $\pm$ 0.045	-0.005 $\pm$ 0.056	0.159 $\pm$ 0.002
SciPlex3	Average effect	LFC	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.084 $\pm$ 0.000
SciPlex3	Average effect	Post	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.084 $\pm$ 0.000
SciPlex3	CPA	LFC	-0.127 $\pm$ 0.978	-0.167 $\pm$ 1.041	0.226 $\pm$ 0.140
SciPlex3	CPA	Post	-0.100 $\pm$ 0.985	-0.083 $\pm$ 1.010	0.228 $\pm$ 0.160
SciPlex3	CPA End-to-End	Embedding	-0.547 $\pm$ 0.476	-0.622 $\pm$ 0.543	0.255 $\pm$ 0.259
SciPlex3	Measured	LFC	0.974 $\pm$ 0.000	1.000 $\pm$ 0.000	0.304 $\pm$ 0.000
SciPlex3	Measured	Post	0.666 $\pm$ 0.000	0.500 $\pm$ 0.000	0.425 $\pm$ 0.000
SciPlex3	Measured	Pre	0.373 $\pm$ 0.000	0.500 $\pm$ 0.000	0.452 $\pm$ 0.000
SciPlex3	Measured	Pre+SMILES	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.458 $\pm$ 0.000
SciPlex3	No effect	LFC	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.084 $\pm$ 0.000
SciPlex3	No effect	Post	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.084 $\pm$ 0.000
SciPlex3	PRnet	LFC	0.052 $\pm$ 0.579	-0.228 $\pm$ 0.637	0.246 $\pm$ 0.150
SciPlex3	PRnet	Post	-0.074 $\pm$ 0.666	-0.228 $\pm$ 0.637	0.247 $\pm$ 0.149
SciPlex3	chemCPA	LFC	0.070 $\pm$ 0.033	0.061 $\pm$ 0.058	0.148 $\pm$ 0.015
SciPlex3	chemCPA	Post	0.071 $\pm$ 0.068	0.069 $\pm$ 0.062	0.152 $\pm$ 0.013

Table 4: Paired fold comparisons vs baselines (within-drug metrics (averaged to fold level); CPA family; best mean diff / lowest FDR p per dataset $\times$ metric in bold).

Dataset	Metric	Source	Model	Baseline	Mean diff [95% CI]	p (FDR)	n
McFarland	pearson	CPA	LFC	Average effect	0.229 [0.023, 0.435]	0.0628	5
McFarland	pearson	CPA	LFC	No effect	0.282 [0.149, 0.414]	0.0247	5
McFarland	pearson	CPA	Post	Average effect	0.238 [0.036, 0.439]	0.0615	5
McFarland	pearson	CPA	Post	No effect	0.239 [0.042, 0.436]	0.0615	5
McFarland	pearson	PRnet	LFC	Average effect	0.032 [-0.097, 0.161]	0.629	5
McFarland	pearson	PRnet	LFC	No effect	0.085 [0.019, 0.151]	0.0615	5
McFarland	pearson	PRnet	Post	Average effect	0.020 [-0.020, 0.060]	0.304	5
McFarland	pearson	PRnet	Post	No effect	0.022 [-0.016, 0.060]	0.272	5
McFarland	pearson	chemCPA	LFC	Average effect	0.245 [0.075, 0.414]	0.0615	5
McFarland	pearson	chemCPA	LFC	No effect	0.297 [0.222, 0.372]	0.00469	5
McFarland	pearson	chemCPA	Post	Average effect	-0.015 [-0.152, 0.122]	0.803	5
McFarland	pearson	chemCPA	Post	No effect	-0.013 [-0.154, 0.127]	0.803	5
McFarland	rmse	CPA	LFC	Average effect	-0.020 [-0.027, -0.013]	0.00769	5
McFarland	rmse	CPA	LFC	No effect	-0.018 [-0.028, -0.007]	0.023	5
McFarland	rmse	CPA	Post	Average effect	-0.014 [-0.022, -0.007]	0.023	5
McFarland	rmse	CPA	Post	No effect	-0.014 [-0.022, -0.006]	0.023	5
McFarland	rmse	PRnet	LFC	Average effect	-0.008 [-0.015, -0.001]	0.0663	5
McFarland	rmse	PRnet	LFC	No effect	-0.006 [-0.006, -0.005]	0.00043	5
McFarland	rmse	PRnet	Post	Average effect	0.054 [-0.133, 0.242]	0.56	5
McFarland	rmse	PRnet	Post	No effect	0.055 [-0.133, 0.242]	0.56	5
McFarland	rmse	chemCPA	LFC	Average effect	-0.009 [-0.019, 0.001]	0.102	5
McFarland	rmse	chemCPA	LFC	No effect	-0.007 [-0.019, 0.005]	0.26	5
McFarland	rmse	chemCPA	Post	Average effect	-0.000 [-0.010, 0.009]	0.967	5
McFarland	rmse	chemCPA	Post	No effect	0.000 [-0.010, 0.010]	0.967	5
McFarland	spearman	CPA	LFC	Average effect	0.163 [-0.002, 0.328]	0.088	5
McFarland	spearman	CPA	LFC	No effect	0.221 [0.122, 0.321]	0.0141	5
McFarland	spearman	CPA	Post	Average effect	0.172 [0.032, 0.312]	0.054	5
McFarland	spearman	CPA	Post	No effect	0.173 [0.037, 0.310]	0.054	5
McFarland	spearman	PRnet	LFC	Average effect	0.014 [-0.077, 0.106]	0.78	5
McFarland	spearman	PRnet	LFC	No effect	0.072 [0.049, 0.095]	0.00584	5
McFarland	spearman	PRnet	Post	Average effect	0.007 [-0.058, 0.071]	0.78	5
McFarland	spearman	PRnet	Post	No effect	0.008 [-0.051, 0.068]	0.78	5
McFarland	spearman	chemCPA	LFC	Average effect	0.223 [0.065, 0.382]	0.0523	5
McFarland	spearman	chemCPA	LFC	No effect	0.281 [0.200, 0.363]	0.00584	5
McFarland	spearman	chemCPA	Post	Average effect	-0.055 [-0.198, 0.088]	0.468	5
McFarland	spearman	chemCPA	Post	No effect	-0.054 [-0.196, 0.088]	0.468	5
SciPlex3	pearson	CPA	LFC	Average effect	—	—	1
SciPlex3	pearson	CPA	LFC	No effect	—	—	1
SciPlex3	pearson	CPA	Post	Average effect	—	—	1
SciPlex3	pearson	CPA	Post	No effect	—	—	1
SciPlex3	pearson	PRnet	LFC	Average effect	—	—	1
SciPlex3	pearson	PRnet	LFC	No effect	—	—	1
SciPlex3	pearson	PRnet	Post	Average effect	—	—	1
SciPlex3	pearson	PRnet	Post	No effect	—	—	1
SciPlex3	pearson	chemCPA	LFC	Average effect	—	—	1
SciPlex3	pearson	chemCPA	LFC	No effect	—	—	1
SciPlex3	pearson	chemCPA	Post	Average effect	—	—	1
SciPlex3	pearson	chemCPA	Post	No effect	—	—	1
SciPlex3	rmse	CPA	LFC	Average effect	—	—	1
SciPlex3	rmse	CPA	LFC	No effect	—	—	1
SciPlex3	rmse	CPA	Post	Average effect	—	—	1
SciPlex3	rmse	CPA	Post	No effect	—	—	1
SciPlex3	rmse	PRnet	LFC	Average effect	—	—	1

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Table 4: Paired fold comparisons vs baselines (within-drug metrics (averaged to fold level); CPA family; best mean diff / lowest FDR p per dataset×metric in bold). (continued)

Dataset	Metric	Source	Model	Baseline	Mean diff [95% CI]	p (FDR)	n
SciPlex3	rmse	PRnet	LFC	No effect	–	–	1
SciPlex3	rmse	PRnet	Post	Average effect	–	–	1
SciPlex3	rmse	PRnet	Post	No effect	–	–	1
SciPlex3	rmse	chemCPA	LFC	Average effect	–	–	1
SciPlex3	rmse	chemCPA	LFC	No effect	–	–	1
SciPlex3	rmse	chemCPA	Post	Average effect	–	–	1
SciPlex3	rmse	chemCPA	Post	No effect	–	–	1
SciPlex3	spearman	CPA	LFC	Average effect	–	–	1
SciPlex3	spearman	CPA	LFC	No effect	–	–	1
SciPlex3	spearman	CPA	Post	Average effect	–	–	1
SciPlex3	spearman	CPA	Post	No effect	–	–	1
SciPlex3	spearman	PRnet	LFC	Average effect	–	–	1
SciPlex3	spearman	PRnet	LFC	No effect	–	–	1
SciPlex3	spearman	PRnet	Post	Average effect	–	–	1
SciPlex3	spearman	PRnet	Post	No effect	–	–	1
SciPlex3	spearman	chemCPA	LFC	Average effect	–	–	1
SciPlex3	spearman	chemCPA	LFC	No effect	–	–	1
SciPlex3	spearman	chemCPA	Post	Average effect	–	–	1
SciPlex3	spearman	chemCPA	Post	No effect	–	–	1

Table 5: Within-cell-line metrics (averaged to fold level) (CPA family; mean $\pm$ s.d. across folds; best per dataset in bold).

Dataset	Source	Model	Pearson r	Spearman ρ	RMSE
McFarland	Average effect	LFC	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.158 $\pm$ 0.037
McFarland	Average effect	Post	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.157 $\pm$ 0.031
McFarland	CPA	LFC	0.496 $\pm$ 0.328	0.393 $\pm$ 0.308	0.130 $\pm$ 0.041
McFarland	CPA	Post	0.450 $\pm$ 0.327	0.379 $\pm$ 0.325	0.140 $\pm$ 0.035
McFarland	CPA End-to-End	Embedding	0.628 $\pm$ 0.194	0.542 $\pm$ 0.216	0.167 $\pm$ 0.029
McFarland	Measured	LFC	0.526 $\pm$ 0.326	0.461 $\pm$ 0.250	0.117 $\pm$ 0.041
McFarland	Measured	Post	0.546 $\pm$ 0.300	0.449 $\pm$ 0.183	0.132 $\pm$ 0.031
McFarland	Measured	Pre	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.153 $\pm$ 0.038
McFarland	Measured	Pre+SMILES	0.310 $\pm$ 0.171	0.263 $\pm$ 0.210	0.142 $\pm$ 0.025
McFarland	No effect	LFC	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.152 $\pm$ 0.036
McFarland	No effect	Post	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.158 $\pm$ 0.031
McFarland	PRnet	LFC	0.372 $\pm$ 0.379	0.315 $\pm$ 0.325	0.147 $\pm$ 0.031
McFarland	PRnet	Post	0.383 $\pm$ 0.396	0.334 $\pm$ 0.327	0.171 $\pm$ 0.100
McFarland	chemCPA	LFC	0.477 $\pm$ 0.042	0.356 $\pm$ 0.036	0.151 $\pm$ 0.001
McFarland	chemCPA	Post	0.439 $\pm$ 0.115	0.316 $\pm$ 0.097	0.159 $\pm$ 0.003
SciPlex3	Average effect	LFC	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.141 $\pm$ 0.025
SciPlex3	Average effect	Post	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.141 $\pm$ 0.025
SciPlex3	CPA	LFC	0.514 $\pm$ 0.152	0.481 $\pm$ 0.130	0.158 $\pm$ 0.013
SciPlex3	CPA	Post	0.540 $\pm$ 0.160	0.480 $\pm$ 0.129	0.156 $\pm$ 0.013
SciPlex3	CPA End-to-End	Embedding	0.391 $\pm$ 0.108	0.346 $\pm$ 0.146	0.226 $\pm$ 0.019
SciPlex3	Measured	LFC	0.426 $\pm$ 0.213	0.320 $\pm$ 0.201	0.133 $\pm$ 0.027
SciPlex3	Measured	Post	0.512 $\pm$ 0.254	0.361 $\pm$ 0.260	0.142 $\pm$ 0.034
SciPlex3	Measured	Pre	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.169 $\pm$ 0.026
SciPlex3	Measured	Pre+SMILES	0.281 $\pm$ 0.268	0.198 $\pm$ 0.182	0.167 $\pm$ 0.025
SciPlex3	No effect	LFC	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.141 $\pm$ 0.025
SciPlex3	No effect	Post	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.141 $\pm$ 0.025
SciPlex3	PRnet	LFC	0.358 $\pm$ 0.056	0.300 $\pm$ 0.059	0.163 $\pm$ 0.018
SciPlex3	PRnet	Post	0.424 $\pm$ 0.027	0.375 $\pm$ 0.068	0.161 $\pm$ 0.016
SciPlex3	chemCPA	LFC	0.296 $\pm$ 0.129	0.204 $\pm$ 0.144	0.169 $\pm$ 0.010
SciPlex3	chemCPA	Post	0.218 $\pm$ 0.125	0.182 $\pm$ 0.133	0.177 $\pm$ 0.016

Table 6: Paired fold comparisons vs baselines (within-cell-line metrics (averaged to fold level); CPA family; best mean diff / lowest FDR p per dataset×metric in bold).

Dataset	Metric	Source	Model	Baseline	Mean diff [95% CI]	p (FDR)	n
McFarland	pearson	CPA	LFC	Average effect	0.496 [0.089, 0.903]	0.0553	5
McFarland	pearson	CPA	LFC	No effect	0.496 [0.089, 0.903]	0.0553	5
McFarland	pearson	CPA	Post	Average effect	0.450 [0.044, 0.856]	0.0557	5
McFarland	pearson	CPA	Post	No effect	0.450 [0.044, 0.856]	0.0557	5
McFarland	pearson	PRnet	LFC	Average effect	0.372 [-0.098, 0.843]	0.0967	5
McFarland	pearson	PRnet	LFC	No effect	0.372 [-0.098, 0.843]	0.0967	5
McFarland	pearson	PRnet	Post	Average effect	0.383 [-0.109, 0.876]	0.0967	5
McFarland	pearson	PRnet	Post	No effect	0.383 [-0.109, 0.876]	0.0967	5
McFarland	pearson	chemCPA	LFC	Average effect	0.477 [0.425, 0.528]	8.21e-05	5
McFarland	pearson	chemCPA	LFC	No effect	0.477 [0.425, 0.528]	8.21e-05	5
McFarland	pearson	chemCPA	Post	Average effect	0.439 [0.297, 0.582]	0.00308	5
McFarland	pearson	chemCPA	Post	No effect	0.439 [0.297, 0.582]	0.00308	5
McFarland	rmse	CPA	LFC	Average effect	-0.028 [-0.050, -0.007]	0.0635	5
McFarland	rmse	CPA	LFC	No effect	-0.022 [-0.037, -0.007]	0.0571	5
McFarland	rmse	CPA	Post	Average effect	-0.018 [-0.026, -0.009]	0.0331	5
McFarland	rmse	CPA	Post	No effect	-0.018 [-0.028, -0.009]	0.0331	5
McFarland	rmse	PRnet	LFC	Average effect	-0.011 [-0.022, 0.000]	0.129	5
McFarland	rmse	PRnet	LFC	No effect	-0.005 [-0.016, 0.007]	0.646	5
McFarland	rmse	PRnet	Post	Average effect	0.014 [-0.083, 0.110]	0.93	5
McFarland	rmse	PRnet	Post	No effect	0.013 [-0.084, 0.110]	0.93	5
McFarland	rmse	chemCPA	LFC	Average effect	-0.008 [-0.055, 0.039]	0.93	5
McFarland	rmse	chemCPA	LFC	No effect	-0.002 [-0.048, 0.045]	0.93	5
McFarland	rmse	chemCPA	Post	Average effect	0.002 [-0.039, 0.043]	0.93	5
McFarland	rmse	chemCPA	Post	No effect	0.001 [-0.039, 0.042]	0.93	5
McFarland	spearman	CPA	LFC	Average effect	0.393 [0.011, 0.776]	0.0891	5
McFarland	spearman	CPA	LFC	No effect	0.393 [0.011, 0.776]	0.0891	5
McFarland	spearman	CPA	Post	Average effect	0.379 [-0.024, 0.783]	0.0891	5
McFarland	spearman	CPA	Post	No effect	0.379 [-0.024, 0.783]	0.0891	5
McFarland	spearman	PRnet	LFC	Average effect	0.315 [-0.088, 0.718]	0.096	5
McFarland	spearman	PRnet	LFC	No effect	0.315 [-0.088, 0.718]	0.096	5
McFarland	spearman	PRnet	Post	Average effect	0.334 [-0.072, 0.740]	0.096	5
McFarland	spearman	PRnet	Post	No effect	0.334 [-0.072, 0.740]	0.096	5
McFarland	spearman	chemCPA	LFC	Average effect	0.356 [0.311, 0.402]	0.000154	5
McFarland	spearman	chemCPA	LFC	No effect	0.356 [0.311, 0.402]	0.000154	5
McFarland	spearman	chemCPA	Post	Average effect	0.316 [0.196, 0.436]	0.00559	5
McFarland	spearman	chemCPA	Post	No effect	0.316 [0.196, 0.436]	0.00559	5
SciPlex3	pearson	CPA	LFC	Average effect	0.514 [0.325, 0.703]	0.00248	5
SciPlex3	pearson	CPA	LFC	No effect	0.514 [0.325, 0.703]	0.00248	5
SciPlex3	pearson	CPA	Post	Average effect	0.540 [0.342, 0.739]	0.00248	5
SciPlex3	pearson	CPA	Post	No effect	0.540 [0.342, 0.739]	0.00248	5
SciPlex3	pearson	PRnet	LFC	Average effect	0.358 [0.289, 0.427]	0.000409	5
SciPlex3	pearson	PRnet	LFC	No effect	0.358 [0.289, 0.427]	0.000409	5
SciPlex3	pearson	PRnet	Post	Average effect	0.424 [0.391, 0.457]	2.21e-05	5
SciPlex3	pearson	PRnet	Post	No effect	0.424 [0.391, 0.457]	2.21e-05	5
SciPlex3	pearson	chemCPA	LFC	Average effect	0.296 [0.135, 0.457]	0.00832	5
SciPlex3	pearson	chemCPA	LFC	No effect	0.296 [0.135, 0.457]	0.00832	5
SciPlex3	pearson	chemCPA	Post	Average effect	0.218 [0.063, 0.374]	0.0175	5
SciPlex3	pearson	chemCPA	Post	No effect	0.218 [0.063, 0.374]	0.0175	5
SciPlex3	rmse	CPA	LFC	Average effect	0.017 [-0.005, 0.038]	0.117	5
SciPlex3	rmse	CPA	LFC	No effect	0.017 [-0.005, 0.038]	0.117	5
SciPlex3	rmse	CPA	Post	Average effect	0.016 [-0.007, 0.039]	0.13	5
SciPlex3	rmse	CPA	Post	No effect	0.016 [-0.007, 0.039]	0.13	5
SciPlex3	rmse	PRnet	LFC	Average effect	0.023 [0.002, 0.043]	0.0539	5

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Table 6: Paired fold comparisons vs baselines (within-cell-line metrics (averaged to fold level); CPA family; best mean diff / lowest FDR p per dataset $\times$ metric in bold). (continued)

Dataset	Metric	Source	Model	Baseline	Mean diff [95% CI]	p (FDR)	n
SciPlex3	rmse	PRnet	LFC	No effect	0.022 [0.002, 0.042]	0.0539	5
SciPlex3	rmse	PRnet	Post	Average effect	0.020 [0.003, 0.037]	0.0539	5
SciPlex3	rmse	PRnet	Post	No effect	0.020 [0.003, 0.037]	0.0539	5
SciPlex3	rmse	chemCPA	LFC	Average effect	0.028 [0.004, 0.053]	0.0539	5
SciPlex3	rmse	chemCPA	LFC	No effect	0.028 [0.004, 0.052]	0.0539	5
SciPlex3	rmse	chemCPA	Post	Average effect	0.036 [0.015, 0.058]	0.0539	5
SciPlex3	rmse	chemCPA	Post	No effect	0.036 [0.015, 0.058]	0.0539	5
SciPlex3	spearman	CPA	LFC	Average effect	0.481 [0.320, 0.643]	0.00174	5
SciPlex3	spearman	CPA	LFC	No effect	0.481 [0.320, 0.643]	0.00174	5
SciPlex3	spearman	CPA	Post	Average effect	0.480 [0.320, 0.640]	0.00174	5
SciPlex3	spearman	CPA	Post	No effect	0.480 [0.320, 0.640]	0.00174	5
SciPlex3	spearman	PRnet	LFC	Average effect	0.300 [0.226, 0.374]	0.00105	5
SciPlex3	spearman	PRnet	LFC	No effect	0.300 [0.226, 0.374]	0.00105	5
SciPlex3	spearman	PRnet	Post	Average effect	0.375 [0.291, 0.460]	0.00105	5
SciPlex3	spearman	PRnet	Post	No effect	0.375 [0.291, 0.460]	0.00105	5
SciPlex3	spearman	chemCPA	LFC	Average effect	0.204 [0.026, 0.382]	0.0371	5
SciPlex3	spearman	chemCPA	LFC	No effect	0.204 [0.026, 0.382]	0.0371	5
SciPlex3	spearman	chemCPA	Post	Average effect	0.182 [0.018, 0.347]	0.0371	5
SciPlex3	spearman	chemCPA	Post	No effect	0.182 [0.018, 0.347]	0.0371	5